Supporting Information

"Shuttle" in Polysulfide Shuttle: Friend or Foe?

Aashutosh N. Mistry[*] and *Partha P. Mukherjee*[†‡]

School of Mechanical Engineering, Purdue University, West Lafayette, IN 47907, United States

This document explains the essential details of the mathematical description of Li-sulfur cell dynamics with chemical reactions at the anode surface.

S1. Chemical Interactions in the Li-sulfur Cell

A major source of complexity in Li-sulfur chemistry is the presence of multiple chemical and electrochemical reactions. Additionally, sulfur exists in different forms – different polysulfide ions in electrolyte phase and solid species. Given the presence of multiple electrolyte species, transport interactions are also closely intertwined.

The following set of reactions are considered at the cathode:

(chemical) dissolution of solid sulfur:

$$S_{8(l)} \underset{\text{dissolution}}{\overset{\text{preccipitation}}{\rightleftharpoons}} S_{8(s)} \tag{S1}$$

(electrochemical) dissolved sulfur to higher-chain polysulfide:

$$\frac{1}{2}S_8^{2-} \rightleftharpoons \frac{1}{2}S_{8(l)} + e^- \tag{S2}$$

(electrochemical) higher-chain to medium-chain polysulfide:

$$S_4^{2-} \rightleftharpoons \frac{1}{2}S_8^{2-} + e^- \tag{S3}$$

(electrochemical) medium-chain polysulfide electrodeposition:

$$\frac{2}{3}Li_2S_{(s)} \rightleftharpoons \frac{4}{3}Li^+ + \frac{1}{6}S_4^{2-} + e^- \tag{S4}$$

Note that during discharge operation, reactions (S2) to (S4) represent a successive electrochemical reduction of sulfur and are the source of electrochemical energy. If any of these molecules (i.e., $S_{8(l)}, S_8^{2-}, S_4^{2-}$) reach anode, high electropositivity of Li metal can chemically reduce. These chemical reduction events represent a direct exchange of electrons from Li to sulfur species in the

[*] ORCID: 0000 – 0002 – 4359 – 4975
[†] ORCID: 0000 – 0001 – 7900 – 7261
[‡] Email: pmukherjee@purdue.edu

electrolyte. This is different compared to the electrochemical reduction of sulfur at cathode where (electrochemical) oxidation of lithium at anode $\left(Li \rightleftharpoons Li^{+} + e^{-}\right)$ releases an electron in the external circuit that reaches cathode and partakes in electrochemical reduction of sulfur. Thus, the portion of sulfur reduction that takes place given the direct exchange of electrons at Li metal anode, refers to a defect in electrochemical performance. Appropriate reactions are as follows:

(chemical) reduction of dissolved sulfur:

$$Li + \frac{1}{2} S_{8(l)} \xrightarrow{\text{reduction}} Li^{+} + \frac{1}{2} S_8^{2-} \tag{S5}$$

(chemical) reduction of higher-chain polysulfide:

$$Li + \frac{1}{2} S_8^{2-} \xrightarrow{\text{reduction}} Li^{+} + S_4^{2-} \tag{S6}$$

(chemical) reduction of medium-chain polysulfide:

$$Li + \frac{1}{3} Li^{+} + \frac{1}{6} S_4^{2-} \xrightarrow{\text{reduction}} \frac{2}{3} Li_2S_{(s)} \tag{S7}$$

Thus, the following species (ionic and neutral) coexists in the electrolyte phase. The relative proportions of each will not only depend on the reaction rates but also on the transport interactions. The electrolyte transport for such a multispecies electrolyte is to be described by concentrated solution theory[1-2] in order to account for the effects arising from interspecies and interspecies interactions. Authors have recently developed a formalism[2] to describe such an interaction. The appropriate forms of transport laws are $\left(p : Li^{+} \mid s : S_{8(l)} \mid x : S_8^{2-} \mid y : S_4^{2-}\right)$:

Li^{+} *ionic flux*:

$$N_p = -\mathfrak{D}_{pp} \nabla C_p - \mathfrak{D}_{ps} \nabla C_s - \mathfrak{D}_{px} \nabla C_x - \mathfrak{D}_{py} \nabla C_y + t_p \frac{I}{F} + C_p V \tag{S8}$$

$S_{8(l)}$ *species flux*:

$$N_s = -\mathfrak{D}_{sp} \nabla C_p - \mathfrak{D}_{ss} \nabla C_s - \mathfrak{D}_{sx} \nabla C_x - \mathfrak{D}_{sy} \nabla C_y + t_s \frac{I}{F} + C_s V \tag{S9}$$

S_8^{2-} *ionic flux*:

$$N_x = -\mathfrak{D}_{xp} \nabla C_p - \mathfrak{D}_{xs} \nabla C_s - \mathfrak{D}_{xx} \nabla C_x - \mathfrak{D}_{xy} \nabla C_y - t_x \frac{I}{2F} + C_x V \tag{S10}$$

S_4^{2-} *ionic flux*:

$$N_y = -\mathcal{D}_{yp}\nabla C_p - \mathcal{D}_{ys}\nabla C_s - \mathcal{D}_{yx}\nabla C_x - \mathcal{D}_{yy}\nabla C_y - t_y \frac{I}{2F} + C_y V \tag{S11}$$

Ionic current:

$$I = -\kappa\nabla\phi_e - \kappa_p \nabla \ln C_p - \kappa_s \nabla \ln C_s - \kappa_x \nabla \ln C_x - \kappa_y \nabla \ln C_y \tag{S12}$$

Note that the flux of each species is affected by gradients in other species concentrations. Additionally, each of these transport coefficients exhibits dependence on local species concentration. The first term in the ionic current quantifies the migrational current, while the remaining four accounts for the diffusional contributions. Here V is the bulk electrolyte velocity.

Given the local charge neutrality, the concentration of primary salt anion (e.g., $TFSI^-$ in *LiTFSI*) is correlated to other concentration and one does not need to track it separately. Here z_a is the charge number of the anion A^{z_a} in primary salt $Li_{\nu_p}A_{\nu_a}$ and ν_p, ν_a are stoichiometric coefficients.

Local charge neutrality (primary salt anion concentration):

$$C_p + z_a C_a - 2C_x - 2C_y = 0 \tag{S13}$$

S2. Electrochemical Description for the Li-sulfur Cell

Following up from the reaction and transport description presented in the previous section, one can identify the appropriate set of governing equations for the Li-S chemistry[3-4] as follows:

Li^+ *balance*: (S14)

$$\frac{\partial(\varepsilon C_p)}{\partial t} = \nabla\cdot\left(\mathcal{D}_{pp}\frac{\varepsilon}{\tau}\nabla C_p\right) + \nabla\cdot\left(\mathcal{D}_{ps}\frac{\varepsilon}{\tau}\nabla C_s\right) + \nabla\cdot\left(\mathcal{D}_{px}\frac{\varepsilon}{\tau}\nabla C_x\right) + \nabla\cdot\left(\mathcal{D}_{py}\frac{\varepsilon}{\tau}\nabla C_y\right) - t_p\frac{J}{F} + R_p$$

$S_{8(l)}$ *balance*: (S15)

$$\frac{\partial(\varepsilon C_s)}{\partial t} = \nabla\cdot\left(\mathcal{D}_{sp}\frac{\varepsilon}{\tau}\nabla C_p\right) + \nabla\cdot\left(\mathcal{D}_{ss}\frac{\varepsilon}{\tau}\nabla C_s\right) + \nabla\cdot\left(\mathcal{D}_{sx}\frac{\varepsilon}{\tau}\nabla C_x\right) + \nabla\cdot\left(\mathcal{D}_{sy}\frac{\varepsilon}{\tau}\nabla C_y\right) - t_s\frac{J}{F} + R_s$$

S_8^{2-} *balance*: (S16)

$$\frac{\partial(\varepsilon C_x)}{\partial t} = \nabla\cdot\left(\mathcal{D}_{xp}\frac{\varepsilon}{\tau}\nabla C_p\right) + \nabla\cdot\left(\mathcal{D}_{xs}\frac{\varepsilon}{\tau}\nabla C_s\right) + \nabla\cdot\left(\mathcal{D}_{xx}\frac{\varepsilon}{\tau}\nabla C_x\right) + \nabla\cdot\left(\mathcal{D}_{xy}\frac{\varepsilon}{\tau}\nabla C_y\right) + t_x\frac{J}{2F} + R_x$$

S_4^{2-} *balance*: (S17)

$$\frac{\partial\left(\varepsilon C_y\right)}{\partial t}=\nabla\cdot\left(\mathcal{D}_{yp}\frac{\varepsilon}{\tau}\nabla C_p\right)+\nabla\cdot\left(\mathcal{D}_{ys}\frac{\varepsilon}{\tau}\nabla C_s\right)+\nabla\cdot\left(\mathcal{D}_{yx}\frac{\varepsilon}{\tau}\nabla C_x\right)+\nabla\cdot\left(\mathcal{D}_{yy}\frac{\varepsilon}{\tau}\nabla C_y\right)+t_y\frac{J}{2F}+R_y$$

<u>*Ionic current conservation*</u>: (S18)

$$\nabla\cdot\left(\kappa\frac{\varepsilon}{\tau}\nabla\phi_e\right)+\nabla\cdot\left(\kappa_p\frac{\varepsilon}{\tau}\nabla\ln C_p\right)+\nabla\cdot\left(\kappa_s\frac{\varepsilon}{\tau}\nabla\ln C_s\right)+\nabla\cdot\left(\kappa_x\frac{\varepsilon}{\tau}\nabla\ln C_x\right)+\nabla\cdot\left(\kappa_y\frac{\varepsilon}{\tau}\nabla\ln C_y\right)+J=0$$

<u>*Electronic current conservation*</u>: (S19)

$$\nabla\cdot\left(\sigma^{eff}\nabla\phi_s\right)=J$$

$S_{8(s)}$ <u>*evolution*</u>: (S20)

$$\frac{\partial\varepsilon_S}{\partial t}=\tilde{V}_S R_1$$

$Li_2S_{(s)}$ <u>*evolution*</u>: (S21)

$$\frac{\partial\varepsilon_{Li_2S}}{\partial t}=-\tilde{V}_{Li_2S}\frac{2J_4}{3F}$$

<u>*Porosity evolution*</u>: (S22)

$$\frac{\partial}{\partial t}\left(\varepsilon+\varepsilon_S+\varepsilon_{Li_2S}\right)=0$$

Note that net local electrochemical reactions are $J=J_2+J_3+J_4$ where subscripts identify the appropriate reaction prescribed earlier. The generation terms for electrolyte species balance (i.e., R's) are correlated to individual reaction rates (R: chemical reaction and J: electrochemical reaction) via reaction stoichiometry. Cathode microstructure evolves in response to precipitation/dissolution events (reactions (S1) and (S4)) and this evolution is accounted for by appropriately evolving the relevant effective properties (tortuosity, active area, conductivity) based on pore-scale calculations reported earlier[3].

Note that the chemical reactions at the anode surface appear as interface fluxes and are incorporated in anode-separator boundary conditions:

Li^+ : $$N_p\big|_{a/s}=\frac{J_{app}}{F}+R_5+R_6-\frac{1}{3}R_7$$ (S23)

$S_{8(l)}$: $$N_s\big|_{a/s}=-\frac{1}{2}R_5$$ (S24)

S_8^{2-} :
$$N_x\big|_{a/s} = \frac{1}{2}R_5 - \frac{1}{2}R_6 \tag{S25}$$

S_4^{2-} :
$$N_y\big|_{a/s} = R_6 - \frac{1}{6}R_7 \tag{S26}$$

Here the applied current density, J_{app} is related to the cell capacity, Q_{cell} via C-rate:

$$J_{app} = \text{C-rate} \cdot Q_{cell} \tag{S27}$$

$$Q_{cell} = \frac{16F\varepsilon_S^0 L_{cat}}{3600\tilde{V}_S} \quad [\text{Ah}] \tag{S28}$$

It is expected that each sulfur species exhibits different reactivity towards Li metal anode (same as they have different exchange current densities for electrochemical reactions at the cathode). For consistency, reaction rates for chemical reduction at the anode are expressed in terms of corresponding electrochemical reaction at the cathode. For example, reaction current density for reaction (S2) is defined as:

$$j_2 = \frac{J_2}{a} = i_2^0 \left\{ \left(\frac{C_x}{C_x^{ref}} \right)^{1/2} e^{\frac{F}{2RT}(\phi_s - \phi_e - U_2)} - \left(\frac{C_s}{C_s^{ref}} \right)^{1/2} e^{-\frac{F}{2RT}(\phi_s - \phi_e - U_2)} \right\} \tag{S29}$$

and equivalently the rate of sulfur reduction at anode surface (reaction (5)) is:

$$R_5 = k_5 \left(\frac{C_s}{C_s^{ref}} \right)^{1/2} \tag{S30}$$

where the rate constant has been expressed in the form of exchange current density and chemical overpotential:

$$k_5 = \frac{i_2^0}{F} e^{-\frac{F\eta_{ch}}{2RT}} \tag{S31}$$

Smaller values of chemical overpotential mean faster kinetics at the anode surface. Here a is a local electrochemically active area (m^2/m^3), j_2 is reaction rate in A/m^2 of the active surface and R_5 is reaction rate in $mol/m^2/s$.

S3. Sulfur Balance Sheet

Sulfur is present in various different forms in the Li-sulfur cell, for example, as solid sulfur impregnated in the carbon cathode or as medium-chain polysulfide in the electrolyte. Since sulfur does not leave the system, the net amount of sulfur must stay constant. In other words, sulfur changes the physical form (solid or dissolved) but such that globally total sulfur content is time invariant. Initially sulfur is present as solid impregnated sulfur, hence initial sulfur content is:

$$\$\left(t=0\right)=\int\limits_{L_{cat}}\frac{\varepsilon_S}{\tilde{V}_S}\ \mathrm{d}x=\frac{\varepsilon_S^0}{\tilde{V}_s}L_{cat} \tag{S32}$$

At any later time, sulfur content can be quantified as:

$$\$\left(t\right)=\int\limits_{L_{cat}}\frac{\varepsilon_S}{\tilde{V}_S}\ \mathrm{d}x+\int\limits_{L_{sep}+L_{cat}}\left(\varepsilon C_s+\varepsilon C_x+\frac{1}{2}\varepsilon C_y\right)\ \mathrm{d}x+\int\limits_{L_{cat}}\frac{\varepsilon_{Li_2S}}{8\tilde{V}_{Li_2S}}\ \mathrm{d}x+\frac{\varphi_{Li_2S}}{8\tilde{V}_{Li_2S}} \tag{S33}$$

Here φ_{Li_2S} is the volume of Li_2S deposited on the anode (m^3 of Li_2S/m^2 of anode surface) due to chemical reduction of medium-chain polysulfide (reaction (S7)). Except for formation at the anode, all the other interactions return the sulfur back to the reaction zone. In other words, φ_{Li_2S} refers to the irreversible sulfur loss from the inventory.

$$\frac{d\varphi_{Li_2S}}{dt}=\tilde{V}_{Li_2S}\frac{2R_7}{3} \tag{S34}$$

Each of these forms of sulfur can be expressed as a fraction of the total sulfur to fix the identity of sulfur at a given stage during electrochemical operation, for example, a fraction of dissolved sulfur is:

$$f_{dissolved-sulfur}=\frac{1}{\left(L_{cat}\varepsilon_S^0 / \tilde{V}_s\right)}\int\limits_{L_{sep}+L_{cat}}\varepsilon C_s\ \mathrm{d}x \tag{S35}$$

S4. Physicochemical Interactions in the Separator

There are two distinct interactions associated with the polysulfide shuttle effect:

i. transport of sulfur species from the cathode
ii. chemical reduction of sulfur at the anode surface

The transport of these species from the cathode is identified by fluxes at the cathode-separator interface, while the chemical reactions take place at anode-separator interface. Separator behaves as a host space for these interactions. Integrating sulfur species balances (expressions (S15) to (S17)) over the separator, one arrives at the following set of identities:

$$\frac{\partial}{\partial t}\left(\int\limits_{L_{sep}}\varepsilon C_s\ \mathrm{d}x\right)=N_s\big|_{a/s}-N_s\big|_{c/s} \tag{S36}$$

$$\frac{\partial}{\partial t}\left(\int_{L_{sep}} \varepsilon C_x \ \mathrm{d}x\right) = N_x\big|_{a/s} - N_x\big|_{c/s} \tag{S37}$$

$$\frac{\partial}{\partial t}\left(\int_{L_{sep}} \varepsilon C_y \ \mathrm{d}x\right) = N_y\big|_{a/s} - N_y\big|_{c/s} \tag{S38}$$

where the fluxes at the anode – separator interface are related to chemical reaction rates as per equations (S24) – (S26), while the ones at the cathode – separator interface are quantified from generalized transport laws (S9) – (S11). Note that the cathode – separator fluxes represent the amount of each of these reactants that are shuttled from cathode to anode. Any mismatch between the two manifests as sulfur accumulation in the separator.

Separator species balances (S36) – (S38) allow a consistent interpretation of the severity of chemical reduction at the anode surface. It also points to the different effects that jointly give rise to a loss in electrochemical performance: shuttling of ions (c/s) and subsequent reactions (a/s). Not necessarily all that shuttles from cathode participate in these side reactions at the anode. For ease of interpretation (across C-rate especially), each of these fluxes is normalized using the applied current density:

$$N_f^* = \frac{FN_f}{J_{app}} \tag{S39}$$

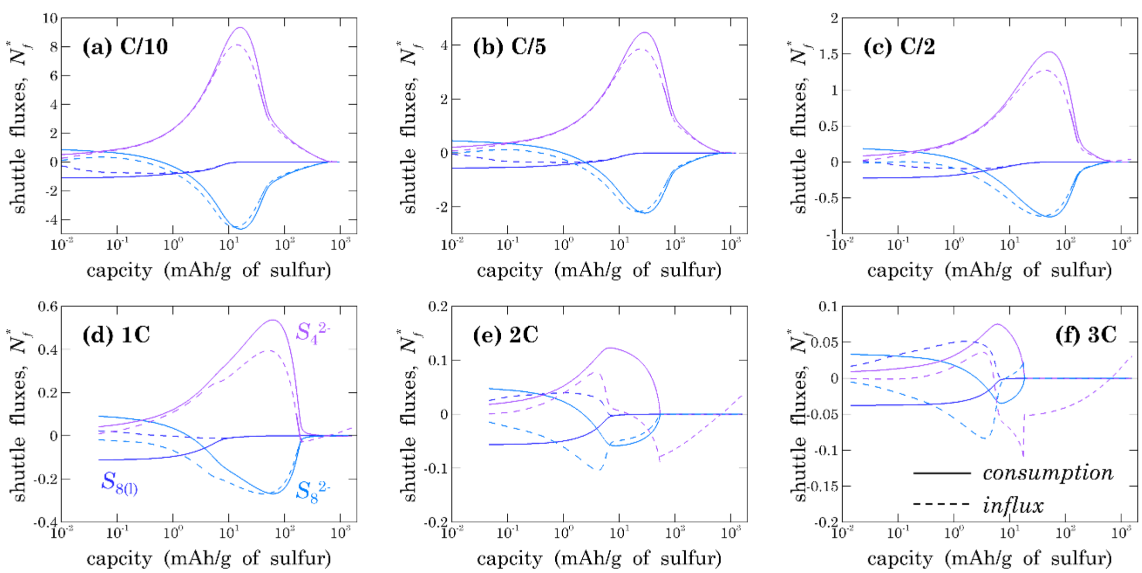

Figure S1: Evolution of sulfur shuttle fluxes change qualitatively as electrochemical reactions are carried out faster.

In response to physicochemical changes taking place inside the cell, sulfur fluxes at anode/separator interface (consumption) and cathode/separator interface (influx) undergo temporal evolution. Figure S1 investigates such a progression as a function of operating rate, while Figure S2 explores the effect of cathode specifications.

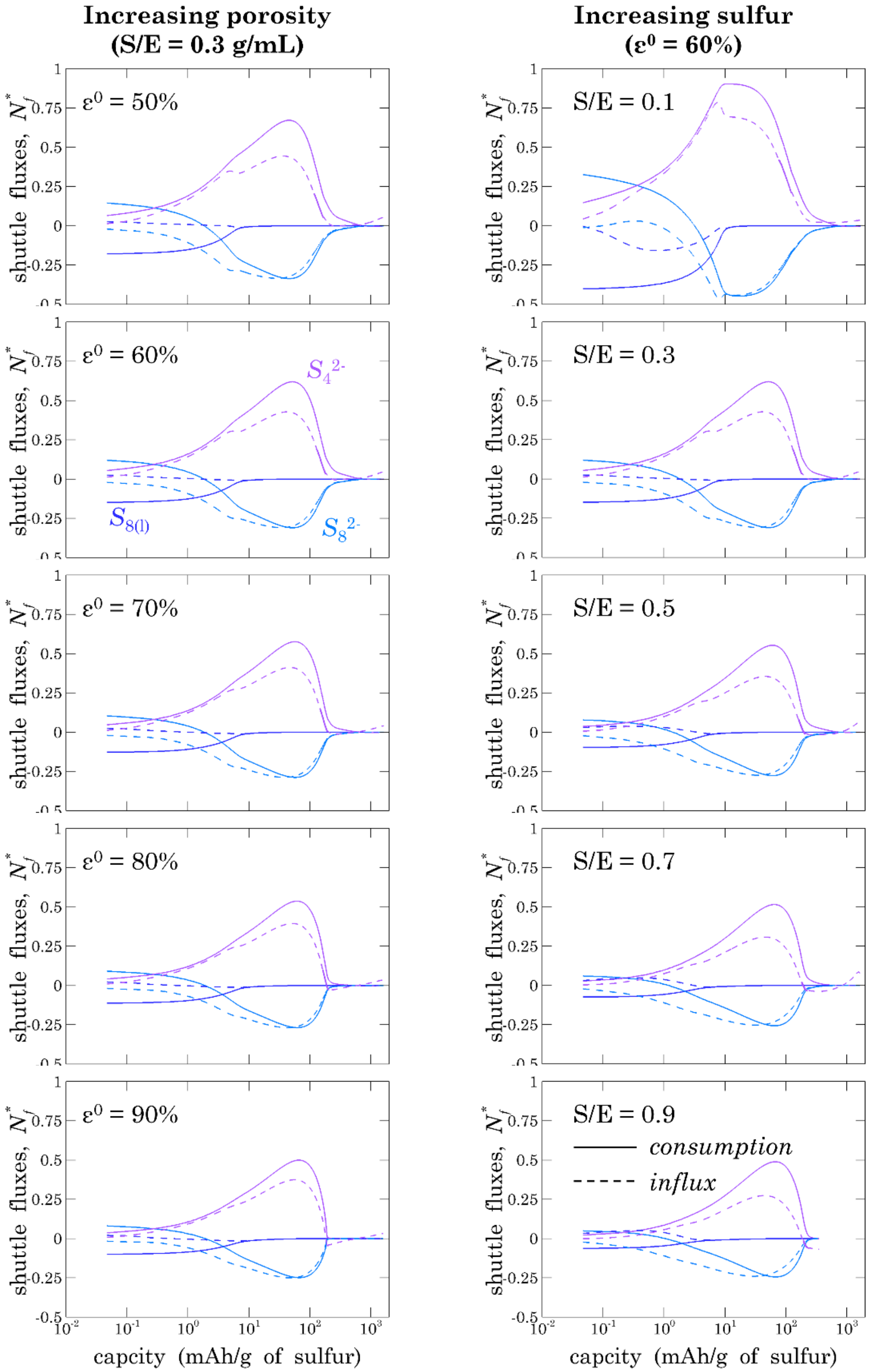

Figure S2: Evolution of polysulfide fluxes is sensitive to cathode specifications, namely, sulfur-to-electrolyte (S/E) ratio and pristine porosity, ε^0.

References

(1) Newman, J.;Thomas-Alyea, K. E. *Electrochemical Systems*; John Wiley & Sons: New Jersey, 2012.
(2) Mistry, A. N.;Mukherjee, P. P. Electrolyte Transport Evolution Dynamics in Lithium–Sulfur Batteries. *J. Phys. Chem. C* **2018**, *122*, 18329-18335.
(3) Mistry, A.;Mukherjee, P. P. Precipitation–Microstructure Interactions in the Li-Sulfur Battery Electrode. *J. Phys. Chem. C* **2017**, *121*, 26256-26264.
(4) Chen, C.-F.; Mistry, A.;Mukherjee, P. P. Probing Impedance and Microstructure Evolution in Lithium-Sulfur Battery Electrodes. *J. Phys. Chem. C* **2017**, *121*, 21206 - 21216.